29/4/23

Module - 5

Hypothesis Testing - I

Level of Configurence = No error $n \geq 30$ large I, Z test $n < 30$ small II, t test

Level of Significance = Error

(1) **Single proportion:-**

Test Statistic (or) Calculated value $\boxed{Z = \dfrac{p - P}{\sqrt{\dfrac{PQ}{n}}}}$

proportion ≈ probability

where $Q = 1 - P$.

P = population proportion

p = Sample proportion

n = sample size

* The CEO of a large electric utility claims that 80 percent of his 10,00,000 customers are very satisfied with service they recieve. To ~~random~~ test this claim, this local newspaper surveyed 100 customers, using simple random sampling. Among the sampled customers, 73 percent say they are very satisfied. Based on these findings, can we reject the CEO's hypothesis that 80 percent of the customers are very satisfied? Use a 0.05 level of significance.

Right tailed = + x

Left tailed = - x

1) Hypothesis : Null $H_0 : P \geq 80\% \Rightarrow P \geq 0.80$

Alternate $H_1 : P < 0.80$

2) Left tailed Test:-

$\alpha = 0.05$, $Z_\alpha = Z_{tab} = -1.64$.

3) $P = 0.80$, $p = 0.73$, $n = 100$

$Q = 1 - P = 1 - 0.80 = 0.2$

$$Z_{cal} = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.73 - 0.80}{\sqrt{\frac{0.80 \times 0.2}{100}}} = \frac{-0.07}{0.004} = -1.75$$

$\boxed{\therefore Z_{cal} = -1.75}$

4) $|Z_{cal}| > |Z_{tab}|$ Accept H_1 ✓
Reject H_0

* Experience has showen that 20% of manufactured product is of top quality. In one day's production 400% particles, only 50 are of top quality - show that either the production of day chosen was not represent. sample or the hypothesis 20% was strong.

Sol^n ① Hypothesis : Null H_0 : 0.20
Alternate H_1: $P \neq 0.20$

$\alpha = 5\%$ (default) if not given

② Two tailed test
$\alpha = 0.05$, $Z_\alpha = Z_{tab} = 1.96$.

③ $P = 0.20$, $p = 0.125$, $n = 400$

$$Z_{cal} = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad ; Q = 1 - P$$

$$= \frac{0.125 - 0.20}{\sqrt{\frac{0.20 \times 0.80}{400}}} = -3.75$$

$$\boxed{Z_{cal} = -3.75}$$

If $|Z_{cal}| < |Z_{tab}|$
Accept H_0
Reject H_1

④ $|Z_{cal}| > |Z_{tab}|$ Accept H_1 ✓
Reject H_0.

* Two proportions:-

Test Statistic $$\boxed{Z = \frac{(P_1 - P_2)}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}}$$

where $P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$, $Q = 1 - P$.

* Suppose the RK Drug Company develops new drug, designed to prevent covid 19. The company states that drug is more effective for women than for men. To test this claim, they choose a simple random sample of 100 women and 200 men from a population of 100,000 volunteers.
At the end of study, 38% of women caught a covid 19 and 51% of men caught covid 19. Based on these findings, can we conclude that drug is more effective for women than for men? Use 0.01 level significance.

Sol:
W > M
(100) (200)
38% 51%

$p_1 = 0.38$, $n_1 = 100$
$p_2 = 0.51$, $n_2 = 200$

$\alpha = 0.01$
left ⇒ 0.49

(i) Hypothesis : Null $H_0 : P_1 \geq P_2$
Alternate $H_1 : P_1 < P_2$

(ii) left tailed test; $\alpha = 0.01$

$Z_{tab} = -2.33$

(iii) $$Z_{cal} = \frac{0.38 - 0.51}{\sqrt{(0.466)(0.534)\left(\frac{1}{100}+\frac{1}{200}\right)}}$$

$$= \frac{-0.13}{\sqrt{0.248(0.015)}}$$

$$= -2.129$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$
$$= \frac{100(0.38) + 200(0.51)}{100+200}$$
$$= \frac{102+38}{300}$$
$$= \frac{140}{300}$$
$$= 0.466$$

$Q = 1 - P = 1 - 0.466$
$= 0.534$.

(iv) $|Z_{cal}| < |Z_{tab}|$
Accept H_0 ✓
Reject H_1.

* In a large city A, 20% of random sample of 900 school days had a slight physical defect. In another large city B, 18.5% of random sample of 1600 school boys had same defect. Is the difference b/w the proportions significant? Use the level of significance 5%.

* Ex.

Sol^n $p_1 = 0.20$, $n_1 = 900$

$p_2 = 0.185$, $n_2 = 1600$.

(i) Hypothesis: $H_0 : P_1 = P_2$

$H_1 : P_1 \neq P_2$

(ii) Two tailed test, $\alpha = 0.05$

$Z_{tab} = 1.96$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$= \frac{900(0.20) + (1600)(0.185)}{2500}$$

$$= 0.1904$$

$$Q = 1 - P = 1 - 0.1904 = 0.8096$$

(iii) $$Z_{cal} = \frac{(0.20 - 0.185)}{\sqrt{(0.1904)(0.8096)\left(\frac{1}{900} + \frac{1}{1600}\right)}}$$

$$= 0.9169$$

(iv) $|Z_{cal}| < |Z_{tab}|$

Accept H_0 ✓

Reject H_1

(iii) Single Sample Mean:-

Test statistic $$\boxed{Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}}$$

$\sigma = s$

$\bar{x}$ = Sample mean

n = sample size

μ = population mean

σ = population S.D.

Note: If σ is not known, sample SD 's' can be used as 's' tends to σ when n is sufficiently large.

* An insurance agent has claimed that average age of policy holders who insure through him is less than average for all agents which is 30.5 years. A random sample of 100 policy holders who had insured through him reveal that mean and S.D. are 28.8 years and 6.35 years respectively. Test his claim at 5% level of significance.

Solr $\mu = 30.5$, $n = 100$, $\bar{x} = 28.8$, σ (or) $s = 6.35$
$\alpha = 0.05$.

(i) Hypothesis : Null $H_0 : \mu = 30.5$
Alternate $H_1 : \mu < 30.5$

(ii) left tailed test:- , $\alpha = 0.05$

$Z_{tab} = -1.64$.

(iii) $Z_{cal} = \dfrac{28.8 - 30.5}{\frac{6.35}{\sqrt{100}}} = \dfrac{-1.7}{0.635} = -2.677$ //

$|Z_{cal}| > Z_{tab}$

(iv) Accept H_1 ✓
Reject H_0

* A sample of 100 students is taken from large population. Mean & height in this sample is 160 cm. Can it be reasonbly regarded that, in the population mean height is 165cm, SD is 10cm? LOS is 1%.

Solr $n = 100$, $\bar{x} = 160$, $\mu = 165$, $\sigma = 10$. , $\alpha = 0.01$

(i) Hypothesis $H_0 : \bar{x} = \mu$
$H_1 : \bar{x} \neq \mu$.

(ii) Two tailed test, $\alpha = 0.01$ 0.495

$Z_{tab} = 2.57$

(iii) $Z_{cal} = \dfrac{160 - 165}{\frac{10}{\sqrt{100}}} = -5$

(iv) $|Z_{cal}| > |Z_{tab}|$
Accept H_1 ✓
Reject H_0.

(iv) Two Sample mean:-

Test Statistic $\boxed{Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}}$

74 * Means of two large samples of 1000 & 2000 members are 67.5 inches and 68 inches respectively. Can samples be regarded as drawn from same population of S.D 2.5 inches?

Sol: $n_1 = 1000$, $n_2 = 2000$, $\bar{x}_1 = 67.5$, $\bar{x}_2 = 68$,
$\sigma_1 = 2.5$, $\sigma_2 = 2.5$

(i) Hypothesis: $H_0 = \bar{x}_1 = \bar{x}_2$
$H_1 = \bar{x}_1 \neq \bar{x}_2$

(ii) Two tailed test, $\alpha = 0.05$

$Z_{tab} = 1.96$

(iii) $Z_{cal} = \dfrac{67.5 - 68}{\sqrt{\dfrac{(2.5)^2}{1000} + \dfrac{(2.5)^2}{2000}}} = \dfrac{-0.5}{\sqrt{\dfrac{6.25}{1000} + \dfrac{6.25}{2000}}}$

$= -5.16$ //

(iv) $|Z_{cal}| > |Z_{tab}|$
Accept H_1 ✓
Reject H_0.

76 * Average marks scored by 32 boys is 72 with S.D of 8, while that for 36 girls is 70 with S.D of 6. Test at 1% level of significance whether the boys perform better than girls.

Sol: $n_1 = 32$, $\bar{x}_1 = 72$, $\sigma_1 = 8$, $n_2 = 36$, $\bar{x}_2 = 70$, $\sigma_2 = 6$.
$\alpha = 0.01$.

(i) Hypothesis: $H_0 = \bar{x}_1 = \bar{x}_2$ (or) $\bar{x}_1 \geq \bar{x}_2$
$H_1 = \bar{x}_1 > \bar{x}_2$

(ii) Right ~~Left~~ tailed test, $\alpha = 0.01$
$Z_{tab} = 2.33$

(iii) $Z_{cal} = \dfrac{72 - 70}{\sqrt{\dfrac{(8)^2}{32} + \dfrac{(6)^2}{36}}} = 1.15$

(iv) $|Z_{cal}| < |Z_{tab}|$

Accept H_0

Reject H_1.